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Time lapse structure-from-motion photogrammetry for continuous geomorphic monitoring

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ABSTRACT: Recent advances are made in earth surface reconstruction with high spatial resolution due to SfM photogrammetry. High flexibility of data acquisition and high potential of process automation allows for a significant increase of the temporal resolution, as well, which is especially interesting to assess geomorphic changes. Two case studies are presented where 4D reconstruction is performed to study soil surface changes at 15 seconds intervals: (a) a thunderstorm event is captured at field scale and (b) a rainfall simulation is observed at plot scale. A workflow is introduced for automatic data acquisition and processing including the following approach: data collection, camera calibration and subsequent image correction, template matching to automatically identify ground control points in each image to account for camera movements, 3D reconstruction of each acquisition interval, and finally applying temporal filtering to the resulting surface change models to correct random noise and to increase the reliability of the measurement of signals of change with low intensity. Results reveal surface change detection with cm- to mm-accuracy. Significant soil changes are measured during the events. Ripple and pool sequences become obvious in both case studies. Additionally, roughness changes and hydrostatic effects are apparent along the temporal domain at the plot scale. 4D monitoring with time-lapse SfM photogrammetry enables new insights into geomorphic processes due to a significant increase of temporal resolution. Copyright © 2017 John Wiley & Sons, Ltd.

KEYWORDS: time-lapse; 4D reconstruction; SfM photogrammetry; geomorphic change detection; soil erosion

Introduction

Since the first application of structure-from-motion (SfM) photogrammetry in geo-scientific studies (James and Robson, 2012), its implementation has increased significantly (Carrivick *et al.*, 2016; Eltner *et al.*, 2016a; Smith *et al.*, 2016) leading to an established method to generate three-dimensional (3D) point clouds describing the earth surface with high resolution.

To study geomorphic changes, SfM implies the advantages of being a technique that is fast, flexible and easy to use regarding data acquisition and post-processing, which increases the significance of high-resolution topography to understand earth surface processes (Tarolli, 2014; Passalacqua *et al.*, 2015). The calculation of 3D models from two-dimensional (2D) images utilizes the stereoscopic principle where an object of interest is observed from different perspectives. SfM photogrammetry can be summarized into three main processing steps (e.g. Eltner *et al.*, 2016a; Smith *et al.*, 2016): (1) identification and matching of homologous image points in overlapping images acquired from different positions; (2) bundle adjustment: reconstructing the data acquisition scheme (i.*E. interior* and exterior camera geometry) and a sparse point cloud (i.e. 3D coordinates of the matched image points); (3) densification of the reconstructed point cloud via multi-view stereo-matching (MVS).

Spatial scales from sub-millimetres (e.g. Snapir *et al.*, 2014) to kilometres (e.g. Dietrich, 2016) have been investigated. However, the temporal resolution in SfM is not exploited to its full potential, yet. In multi-temporal case studies (intervals of months and greater; after Eitel *et al.*, 2016) usually solely a few digital elevation models (DEMs) or point clouds representing different frames in time are calculated to describe geomorphic changes.

There are studies, which utilize the time-lapse principle on 2D information to largely increase the temporal resolution. Images captured in a pre-defined interval are used to perform change analysis, e.g. for snow cover monitoring (Parajka *et al.*, 2012), identification of frost creep processes (Matsuoka, 2014), gully headcut retreat and piping observations (Nichols *et al.*, 2016), land slide tracking (Gabrieli *et al.*, 2016), or to monitor wood loads of rivers (Kramer and Wohl, 2014). Martin-Brualla *et al.* (2015) mine internet image collections and achieve time-lapse sequences of *inter alia* geomorphic changes and thus enable continuous process monitoring to a certain degree depending on available image quality. If the monoscopic setup of the image acquisition is designed carefully, even metric measurements with high accuracy are possible, e.g. measuring velocity fields for glaciers (Maas *et al.*, 2013) or recognizing drainage tunnels after glacial lake outburst flood (GLOF) events (Schwalbe *et al.*, 2016).

All these examples underline the importance of continuous process surveillance in geo-scientific research. However, productivity of observed processes in terms of e.g. eroded masses or surface volume changes remains unfeasible because these monoscopic approaches do not benefit from real 3D information, which is possible with a time-lapse setup from several overlapping perspectives. Different reasons can be listed for the so far missing implementation of SfM photogrammetry at a high temporal rate: the need for automatic data processing to compare a large number of DEMs, the pre-requisite of a robust observation setup to allow for reliable multi-temporal data assessment (e.g. considering adjustments of camera implementations; Mosbrucker *et al.*, 2016), and the technological demand required for a precise camera synchronization. To the knowledge of the authors, only James and Robson (2014a) illustrate the potential of topographic data with very high temporal resolution (hyper-temporal according to Eitel *et al.*, 2016). They generate point clouds from images captured at a minute interval by synchronized cameras to monitor lava flows. The 3D reconstruction of dynamic scenes over time [or four-dimensional (4D) photogrammetry] opens a wide field for researchers of earth sciences and exciting applications are to be expected, especially considering the vast amount of potentially new information regarding observable earth surface changes.

Soil erosion studies are one particularly interesting area of application. The importance of increased spatial resolution to better understand soil erosion has already been depicted for studies at field plots (Eltner *et al.*, 2015) and in gully systems (Kaiser *et al.*, 2014; Stöcker *et al.*, 2015) as well as at rainfall simulation plots (Hänsel *et al.*, 2016; Prosdocimi *et al.*, 2017). However, considering temporal resolution soil erosion studies also demand for an increase in this regard due to the complexity of the processes involved (e.g. Morgan, 2005) and due to fast changes occurring during a single precipitation event (e.g. Favis-Mortlock *et al.*, 2000; Darboux *et al.*, 2001). Furthermore, monitoring of changes of event frequency and magnitude are important (Bracken *et al.*, 2015). However, till today quantitative validation of the concepts of soil erosion, at least at plot scale, is still missing (Kinell, 2016).

In this study, an approach is presented aiming for 4D reconstruction of surface changes during two events: a single thunderstorm and an artificial rainfall simulation. Thereby, DEMs are generated from SfM derived point clouds at two different spatial scales and at 10 second intervals during observations lasting at least 1.5 hours. The setup of the image acquisition scheme and subsequent necessary post-processing to enable highly resolved time-lapse SfM photogrammetry is displayed in detail. To reliably identify surface changes accuracy analysis is performed using independent references. Furthermore, the DEMs of difference (DoDs) are filtered via a temporal filter to eliminate data noise. The final datasets of both observed events enable the identification of intra-event changes.

Study Area

Two events of soil erosion due to precipitation and runoff were investigated; one at field scale and another one at plot scale (Figure 1). Both events were measured at a field in the Saxonian loess province situated in eastern Germany, Europe. Soils in this region are vulnerable to erosion due to the low aggregate stability Pesci and Richter 1996. This fragile landscape was chosen intentionally to increase the potential of soil surface changes and thus to better evaluate and assess the suitability and performance of 4D reconstruction for close-range applications.

The soil at the investigated plots can be characterized as a topped Luvisol, with a silty loam texture consisting of 4% sand, 79% silt and 17% clay (Hänsel *et al.*, 2016). Organic carbon content is low with 1.13%. Climatic conditions of the temperate area are characterized by an average annual accumulated rainfall below 600 mm and an average annual temperature below 9°C.

Experiment 1: Field scale

The first erosion study was conducted on 23 May 2016 during a thunderstorm. The observation period lasted for 1.5 hours. The area of interest has a size of about 4 m × 5 m and is located at the bottom of a 75 m long slope. Thus, a position of preferential deposition is investigated. The cumulative precipitation of that event was 37.5 mm, with 26 mm falling during the first hour. Simultaneously to the image acquisition, runoff was measured at the end of the field plot using a trough in combination with tipping buckets of 1 and 3 l capacity. During the observed time frame, total runoff amounts 450 l. Precipitation data was received from RADAR data with 1 km spatial resolution and five minutes temporal resolution (https://kachelmannwetter.com/de/regenradar/mittelsachsen/).

Experiment 2: Plot scale

The second erosion study was performed on 27 May 2016. This time, an artificial rainfall simulator was utilized, described in detail in Schindewolf and Schmidt (2012). Three linked identical rainfall modules were equipped with oscillating VeeJet80/100 nozzles covering an area of 3 m × 1 m. Rainfall intensity was set to 1 mm/min. The rained micro-plot was bounded by metal sheets to capture the entire runoff and sediment yield from the area of interest. At a sediment collector at the lower end runoff was measured and sediment samples were taken. The experiment was divided in to an infiltration and a subsequent runoff experiment. After a steady state of rainfall and runoff was achieved, additional sediment loaded runoff was pumped from a reflux barrel into the plot, during the concentrated runoff experiment. Additional runoff was stopped once to ensure an enhanced capture of morphologic changes during the run.

Suspended sediments were sampled at the bottom outlet of the plot in five minutes intervals during the infiltration experiment and every minute during the increased runoff experiment. Runoff, that was not sampled, was transferred to the runoff reflux barrel. The soil loss is averaged for the area of interest. Final soil loss (in millimetres) was calculated as the sum of sediment yields (in kg/min) divided by the soil initial bulk density of 1.41 g/cm^3 and the plot area of 3 m^2 (more detail in Hänsel *et al.*, 2016).

Methods

A detailed description of data acquisition and processing is given in the following. The entire workflow of the data processing implemented in this study is illustrated in a flowchart (Figure 2).

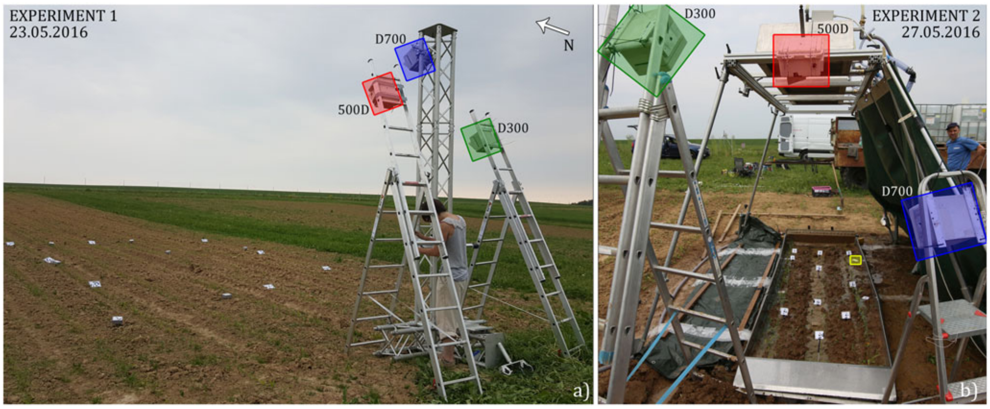

Figure 1. (a) Experimental setup of Experiment 1 at the field plot. (b) Setup of Experiment 2 at the rainfall simulation plot. In both experiments three different SLR cameras are utilized in waterproof housings (green, blue, and red coloured boxes) at heights between 3 and 4 m at the field scale (a) and at a maximum height of about 2 m at the plot scale (b). The yellow box (b) indicates the location of a ground control point (GCP), which is used to illustrate matching performance in Figure 3. [Colour figure can be viewed at wileyonlinelibrary.com]

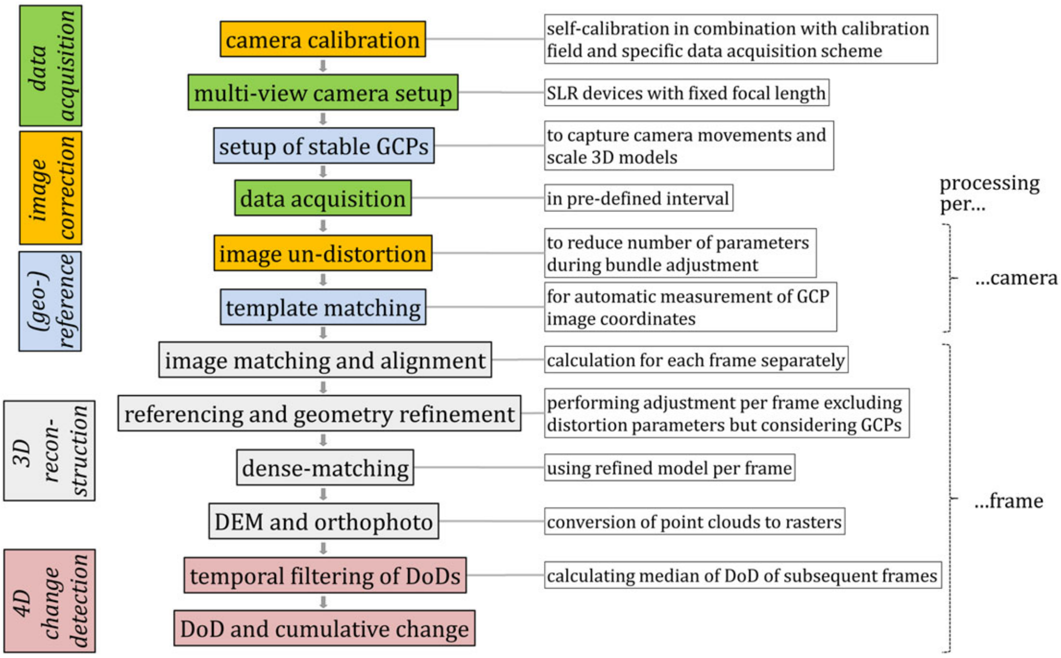

Figure 2. Schematic illustration of the data acquisition and processing chain developed for four-dimensional (4D) surface change reconstruction with structure-from-motion (SfM) photogrammetry. [Colour figure can be viewed at wileyonlinelibrary.com]

Data acquisition

Similar installations with three SLR (single lens reflex) cameras and fixed focal lengths were set up for both erosion studies (Figure 2). The cameras were installed in waterproof housings to protect them from rain, humidity and dust. Images were captured in automatic exposure time mode in order to cope with differing illumination conditions, especially during the thunderstorm. Aperture size was fixed to guarantee sufficient exposure to light and sharpness (f5.6 and f11 for Experiment 1 and 2, respectively). Further camera specifics are listed in Table I.

The images were captured almost simultaneously by each camera. Thereby, automated camera triggering was used. Synchronization was carried out using audio communication between two operators counting to zero to start the camera triggers, which results in temporal offsets below one second. In the following study description, each captured and processed dataset (image triplet) per defined interval will be addressed as frame.

During the thunderstorm event, the cameras were put up above the ground as high as possible to enhance the image acquisition scheme avoiding high incidence angles at the

Table I. Camera specifications of the three different SLR (single lens reflex) cameras used for four-dimensional (4D) reconstruction

	Focal length (mm)	Pixel size (μm)	Sensor resolution (pixel)	Close range ground sampling distance (mm)		f-Number	
				Experiment 1	Experiment 2	Experiment 1	Experiment 2
500D	20	4.3	4752 × 3168	1.1	0.4	5.6	11
D300	28	5.5	4288 × 2848	1.0	0.4	5.6	11
D700	20	8.5	4256 × 2832	2.1	0.9	5.6	11

surface (Eltner *et al.*, 2016b). Therefore, two ladders with an approximate height of 3 m were installed. Furthermore, a 4-m high tripod was used for the third camera. The ladders were fixed to the more stable tripod allowing for better resistance to gusts of wind. Images were taken in 10 seconds intervals, resulting in 581 images per camera. During the rainfall simulation experiment the cameras were fixed to the simulator itself and again to ladders. The highest cameras were situated 2 m above the ground resembling the level of the simulators' nozzles. Triggering interval was set to 10 seconds again, resulting in 589 images per camera.

Image correction

Using SfM and MVS methods to reconstruct the soil surface implies some special considerations in this study. Solely three images are captured, which corresponds close to the minimal requirement of two overlapping images for stereo-matching. Thus, for a reliable surface reconstruction, avoiding occlusion effects and guaranteeing data redundancy to minimize errors, usually significantly more images would be needed. In addition to this circumstance, images are taken by different cameras resulting in a different camera model to be reconstructed for each image. Thus, a robust adjustment approach is needed or else systematic errors, such as dome effects (James and Robson, 2014b; Eltner and Schneider, 2015), can prevent multi-temporal data assessment.

We minimize the parameters to be estimated by bundle block adjustment for camera geometry reconstruction using camera self-calibration prior or posterior the actual experiment to calculate the interior camera parameters. For this procedure, and assuming stable geometry, coded markers were used at a local calibration field. Thereby, mutual location only of a few markers was measured using a measuring tape. Images of these markers were captured handheld in a specific acquisition scheme (e.g. considering rotation and convergence) to avoid parameter correlation. This approach needs information about pixel size, sensor resolution and estimates of focal length but it does not need a precise scale. Estimated interior camera parameters were used to undistort and thus correct the raw images. Thereby, distortion effects including radial-symmetric distortion due to refraction changes, asymmetric and tangential distortion due to lens misalignment, and affine deviations due to non-quadratic pixels were mitigated (Luhmann *et al.*, 2014). Interior camera parameters can be set as fixed during actual scene reconstruction. For more details regarding this approach we refer to Eltner and Schneider (2015). For future studies, a hand-held calibration field can simplify this processing step *in situ*, because in doing so cameras can remain fixed at the installation, which is especially important for long-term observations.

(Geo-)referencing

GCP measurement

Data referencing is a prerequisite for multi-temporal change detection. Thus, 16 GCPs were installed for the thunderstorm event (Experiment 1, Figure 1a). They are wooden circle markers put into marking pipes, which were brought up to 60 cm into the ground (e.g. Eltner *et al.*, 2013), ensuring target stability during the storm. The GCPs were measured with a total station resulting in millimetre accuracy.

For the rainfall simulation (Experiment 2) a different approach is implemented. This time the coded markers (in total 13 GCPs, Figure 1b), used for camera self-calibration, were also employed for referencing. This is possible because during self-calibration, utilizing the specific data acquisition scheme, the coordinates of the coded markers are adjusted, as well (Kaiser *et al.*, submitted for publication). However, in contrast to the sole usage of bundle adjustment to retrieve distortion parameters, marker coordinates need to be known with sufficient precision. We conducted an experiment using scales of superior accuracy confirmed by laser interferometer measurement and total station measurements as references to illustrate the suitability of this approach to retrieve GCP coordinates and to estimate how accurate approximations of markers need to be to achieve corresponding accuracy of GCPs.

Retrieving GCP coordinates via the calibration approach can be performed in two manners. On the one hand, distance measurements can be used to scale the model. Thereby, large distances are mandatory because similar absolute measuring errors at different distances have a corresponding varying influence at the relative error preferring the longer distances to keep the effect at the final model minimal. Afterwards, GCPs are retrieved from the scaled model. On the other hand, GCPs can be measured directly in a local system, e.g. again using a measuring tape. Thereby it is understood that the higher the accuracy of the coordinate estimations the better will be the final GCP accuracy. In both manners of GCP retrievals high-precision scales, which are oriented perpendicular and cover varying distances, can be implemented additionally to improve the performance.

In this study, a GCP setup similar to the rainfall simulation in the field was installed (Figure 3). SLR camera Nikon D700 was used to acquire images. Subsequently, bundle adjustment was performed using different sets of observations; long distances measured via measuring tape and GCP estimates with different added random errors (0.5, 2, and 5 cm), simulating the usage of a measuring tape or another comparable low-accuracy device to retrieve GCP approximations. Furthermore, precision scales are implemented additionally during adjustment to indicate increased accuracy potential using such superior-accuracy tools.

Template matching

In both rainfall events almost 600 images are captured by each camera. Thus, manual measurement of the GCP targets in the

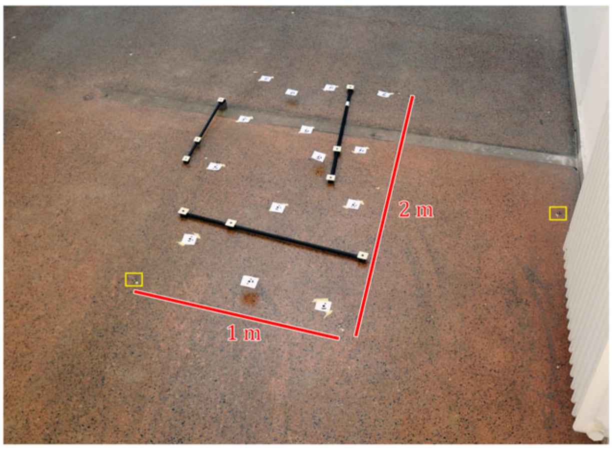

Figure 3. Reference plot (building floor) to measure accuracy potential of oblique multi-camera structure-from-motion (SfM). Coded markers are installed similar to the rainfall simulation experiment (Figure 1b). Referencing similar to the field plot experiment (Figure 1a) is performed with black crosses at the floor (yellow box). Building floor has been measured with total station. Included scales exhibit sub-millimetre accuracy. [Colour figure can be viewed at wileyonlinelibrary.com]

images to reference the data would be impractical because of the very high number of manual measurements needed (581 images × three cameras × 13 targets). Instead, the image coordinates of the GCPs are measured automatically using template matching. Template matching implies the definition of a pattern, in this study defined by the GCP, which will be searched for in the subsequent image. To minimize processing time a maximal search area is defined (between 30 and 200 pixels depending on image resolution and distance to the camera). Therefore, target coordinates (position of matched template) are assigned in one frame and this information is used in the subsequent frame to define the location of search area for the corresponding point in this frame. During actual matching the normalized cross-correlation is calculated to find the location, at which similarity between template and target image are greatest (Figure 4). Thereby, template image T is slid across the source image I resulting in a matrix containing similarity estimates at each pixel location in the source image (Equation (1), after Bradski and Kaehler, 2008). The template matching can be improved to sub-pixel accuracy using least square matching after the initial position of the target has been detected (Grün, 2012). With this approach solely in the first image triplet GCPs need to be assigned manually and in all subsequent frames GCPs are allocated automatically.

$$R(x,y)=\frac{\sum_{x',y'}\left(T(x',y')I(x+x',y+y')\right)}{\sqrt{\sum_{x',y'}T(x',y')^2\sum\limits_{x',y'}I(x+x',y+y')^2}} \qquad (1)$$

Three-dimensional (3D) reconstruction and DEM processing

Automation of data processing is recommended again to calculate soil surface models for about 600 frames. The SfM photogrammetry tool Agisoft PhotoScan allows for such an approach due to an Integrated Development Environment (IDE) using Python programming language. Thus, in this study each frame is processed in the following order. Images of each frame are loaded into an individual chunk (corresponding to a project folder). Afterwards, initial camera specifications, i.e. pixel size and focal length, are assigned to each camera. In a next step image coordinates of GCPs, which were measured via template matching, are read from a corresponding text file and related to the respective GCP object space coordinates. Thereby, also check points are indicated, which are GCPs not considered in the bundle adjustment, for an independent reconstruction performance assessment. After image matching and alignment, a sparse 3D point cloud is reconstructed for each frame. Subsequent densification of these point clouds is performed to receive the final dense point clouds. DEMs are

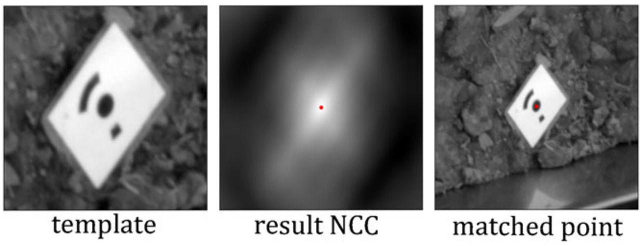

Figure 4. Exemplary illustration of template matching. Left image depicts the template, middle image the result of the normalized cross-correlation (NCC) used to detect the match location of highest correspondence and final matching outcome. The location of the marker is displayed in Figure 2. [Colour figure can be viewed at wileyonlinelibrary.com]

calculated, avoiding interpolation, with a resolution of 10 mm in Experiment 1 and 1.5 mm in Experiment 2. Finally, DEM and images of each frame are used to generate orthophotos. Overall, about 600 DEMs are estimated for each experiment.

Four-dimensional (4D) change detection

The DEMs are subsequently processed for calculating the cumulative DoDs ($DoD_{cumulative}$, Equation (2)) by subtracting the DEM of each frame (DEM_t) from the DEM of the first frame (DEM_0):

$$DoD_{cumulative}=DEM_0-DEM_t \qquad 0\leq t<N \qquad (2)$$

In order to reduce the error in DEM comparison, a non-linear median filtering technique was implemented by averaging the values of *n* Nearest Neighbour (NN) along the temporal domain. Randomly distributed errors can be considerably smoothed due to the high data redundancy contained in the multi-temporal point clouds, allowing to increase the signal-to-noise ratio (SNR) (Abellán *et al.*, 2009; Kromer *et al.*, 2015). NN averaging was originally proposed along the spatial dimension allowing reducing the DoD errors up to a factor of six (Abellán *et al.*, 2009). The median filtering technique was further upgraded to the spatial and temporal dimension, allowing error reduction by a factor of 10 (Kromer *et al.*, 2015). The main idea of the median filter along the temporal domain is to run a one-dimensional (1D) algorithm through DoD and replacing each cell with the median value of neighbouring entries, e.g. computing the median over the first *k* preceding and following entries. This type of filtering technique is specially designed for characteristic speckle noise and 'salt and pepper' noise types (Arce, 2005).

The DoDs of a specific number *k* of consecutive frames (i.e. 10 datasets in our case studies) are stacked together to achieve a 3D matrix of changes. Subsequently, the non-linear filtering technique is calculated along the temporal axis of the matrix. This step is calculated for each pixel of the DoD $-d_{x,y,z}$, as follows:

$$\overline{d}_{x,y,z}=\frac{1}{k}\sum_{i=0}^{k}d_{x,y,z+i} \qquad (3)$$

A spatial filtering was not optimal here because DEMs were already considerably smoothed within the 3D reconstruction chain of PhotoScan after using a mild depth filter, so we prefer using the median filter along the temporal domain of DoDs in our study using Equation (3). The first 20 DoDs of each experiment were used to calculate a 'calibration set' following Kromer *et al.* (2015). This calibration set is a raster similar to $-d_{x,y,z}$, but on which a given number of DoD are stacked ($k=20$ periods in our case study). The resulting smoothed DoD is subtracted from each temporally filtered DoD in order to eliminate systematic errors. This calibration set can be utilized because at the beginning of both experiments no significant changes occur at the soil surface, i.e. for the thunderstorm the first DEMs are captured during precipitation with very low intensity and for the rainfall simulation the first DEMs are acquired when the rainfall simulation has not started.

Accuracy assessment

To assess the accuracy of the reconstructed soil surface independent reference sources are needed. In this study, two different approaches are conducted. The first utilizes the experimental setup at a building floor (already displayed in

Figure 3). The second approach uses SfM photogrammetry to retrieve the 3D description of the soil surface at the field and rainfall simulation. Thereby, the reference is calculated utilizing the traditional SfM setup with more images capturing the area of interest from several perspectives and thus is expected to be more reliable than the models from the image triplet setup.

Building floor reference

The building floor experimental setup has been measured with millimetre-level precision for a study by Eltner and Schneider (2015), aiming to estimate the accuracy of different unmanned aerial vehicle (UAV)-photogrammetry approaches. The floor with a granite texture was captured with total station and mini prism (reflector) measurements at 1 m grid resolution. In this study, the total station measured grid is again used as reference, however using reflector-less mode this time. The time-lapse camera setup of Experiment 1 (field scale) and Experiment 2 (plot scale) is imitated capturing images with the same cameras and lenses at similar relative positions and orientations compared to the setup in the field. The reconstructed models, using distorted and undistorted images, are compared to the total station grid to display accuracy potential of laboratory conditions and to reveal the necessity to undistort the images.

Multi-image ($N > 3$) reference

The accuracy of the 4D surface models from the thunderstorm event are analysed via two models resolving from an UAV campaign prior and another campaign posterior the investigated event. The prior mission has been conducted ten days before the thunderstorm event and the posterior mission five days after the evaluated event. However, in between the prior mission and the 23 May 2016 solely a rainfall event with low intensity occurred (total precipitation amount is 7 mm) and after the thunderstorm event no more rainfall happened until the posterior mission. Thus, changes are expected to be minimal. During both missions about 100 images taken from flying altitudes between 5 and 30 m are processed in PhotoScan to retrieve two point clouds (about 2.5 Mio points each). The internal precision of reconstruction was 1.5 and 3.2 mm for the prior and posterior flight, respectively. The first point cloud of the time-lapse setup is compared to the prior UAV mission point cloud and the last point cloud of the same setup is compared to the posterior UAV mission point cloud.

The rainfall simulation experiment was assessed by terrestrial SfM models. Again, prior and posterior to the actual simulation images were captured walking around the entire plot. The internal precision of reconstruction was 0.4 and 0.9 mm, respectively. About 50 images were acquired during each walk. The point clouds (about 15 Mio points) were reconstructed in PhotoScan and subsequently compared to the similar thunderstorm event, i.e. the first point cloud of the time-lapse setup to the prior mission point cloud and the last point cloud of the time-lapse setup to the posterior mission point cloud. In contrast to the thunderstorm event, no changes compared to the reference data are expected because any time-lag could be avoided.

In both cases (Experiment 1 and 2) the resulting SfM models from multi-images can solely be considered as an approximation of the accuracy because they do not exhibit a superior accuracy. Nevertheless, they can reveal large model deviations due to blunders or systematic errors and they indicate the minimal assumable accuracy potential. Terrestrial laser scanner (TLS) could be a possible independent reference. However, Eltner and Baumgart (2015) show the difficulty of using TLS in soil erosion studies due to surface characteristics (i.e. edge effects at rough surfaces) and data acquisition scheme (i.e. high incidence angles).

Level of detection (LoD) estimation

Although temporal filtering is performed, a level of detection (LoD) is measured, as well, because the magnitude to which the filter is successful has not been tested for the approach in this study. LoD is proposed by Brasington *et al.* (2003) and Lane *et al.* (2003) to receive an error estimate for the DoD. Thereby, error propagation is performed, considering the error of each DEM. In this study, LoD is considered at each temporally filtered DoD to guarantee measured changes are significant. LoD is estimated using the averaged reprojection errors at the control points (CPs) and GCPs of all frames. These error estimates are weighted giving the CPs a higher weight, i.e. the error is considered twice, due to their independence to the adjustment.

Results

Accuracy potential

GCP retrieval

Results, shown in Figure 5, reveal that deviations of estimated GCPs compared to total station measurement are similar for all three chosen error magnitudes of GCP estimates. Furthermore, deviations are high (2.2 mm standard deviation) because accuracy of the total station measurement itself is solely 1.7 mm due to the usage of reflector-less distance measurements. The constant deviation of total station retrieved coordinates to the estimated GCPs of all variants of observation implementations indicates the limited reliability of the total station results, especially when considering the precision scales. The superior accuracy distances display a different picture, clearly highlighting the decreased error with increased accuracy of GCP approximations. Using long distances instead of direct GCP estimations reveals better results (1.4 mm deviation). Furthermore, in future applications precision scales should be included additionally, indicated by an increase of the final accuracy below 0.01 mm, however considering that the scales are included as constraint in the adjustment.

Camera movement compensation

The time-lapse cameras were unsteady during the time span of the study due to movements during gusts of strong wind (Experiment 1) and movements of the oscillating nozzles of the rainfall simulator (Experiment 2). However, movements by the cameras are accounted for in our workflow due to the usage of stable GCPs to reference each frame separately. A closer

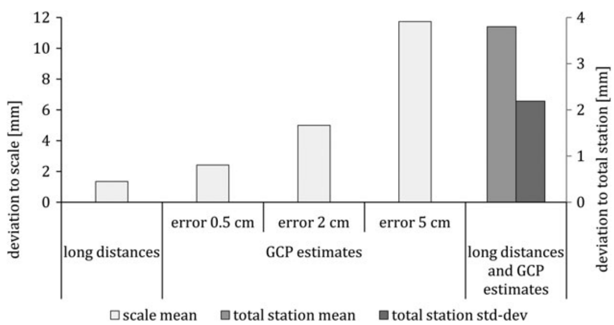

Figure 5. Accuracy of ground control point (GCP) coordinates retrieval using scaled models via long distance measurements and direct GCP estimates from tape measurements. Results are displayed as residuals retrieved from comparison to total station coordinates (mean and standard deviation – std-dev) and to high-precision scales (mean). GCP estimate errors range between 0.5 and 2 cm. Deviations to total station values are similar for all GCP estimate error ranges and the long distance application (consolidated in the last two columns).

look at the camera behaviour during the events reveals their movement depending on the location they are put up at. During the thunderstorm the cameras on the light aluminium ladders are moving several millimetres (average 5 and 13 mm) and a lot stronger compared to the camera on the tripod, which only moved about 0.9 mm. Also, during the rainfall simulation the camera fixed to the simulator shows highest position deviations (average 5.1 mm), which is due to the movement of the oscillating nozzles, whereas the cameras on the more stable ladders moved only 2.5 and 3.9 mm. Overall, the usage of GCPs is needed or cameras have to be installed with higher resistance in cases where no GCPs can be deployed.

Reference comparison – Building floor

The differences between SfM model and building floor are different for distorted and undistorted images (Table II). The corrected images reveal a significantly lower standard deviation compared to the distorted images revealing the benefit of a prior or posterior camera calibration. Higher systematic shifts (mean difference) of the models from the undistorted images are revealed, which are accounted for using iterative closest point (ICP) adjustments. However, these systematic errors are not as relevant for multi-temporal applications if this error is due to imprecise manual GCP image measurements, as in this study, because in each subsequent frame GCPs are assigned at the same position due to automatic matching and thus mitigating the impact. Generally, in both experiment simulations reconstructed surfaces from undistorted images are below 2 mm at the field scale and about 1 mm at the plot scale. Thereby, it needs to be considered that the assessment of the total station grid comprises the disadvantage that a planar surface is assumed between the grid points, which however cannot be ensured. Thus, remaining deviations after ICP application are also attributable to this issue.

Reference comparison – Multi-image

Consulting the reference data to assess the performance of the soil surface reconstruction at the field and rainfall simulation reveals errors of the range of millimetres (Table III). The standard deviation of the calculated point cloud distances between the time-lapse models and the UAV models range from 6 to 10 mm. Consideration of the terrestrial reference highlights even better accuracies due to the closer range to and thus higher resolution of the area of interest (Figure 6). Errors are between 1.4 and 2.1 mm. In both experiments point deviations are further distinguished for the case of ICP implementation. The ICP was used to account for potential remaining systematic errors between the compared models due to alignment issues (e.g. Micheletti *et al.*, 2014), which can mask the assessment of actual performance of surface reconstruction. The ICP consideration decreases the average error for both experiments.

Reprojection error and LoD estimation

The evaluation of the reprojection error of the GCPs and CPs resulting from the adjustment procedure in PhotoScan depicts a higher error compared to the usage of independent references as quality sources (Table IV). Thereby, the error is averaged from all 600 frames of each experiment. CP errors are significantly higher than deviations at the GCPs, which is due to the fact that the GCPs are implemented in the 3D reconstruction and thus deviations are minimized at these points during the adjustment, which is not the case for the CPs.

For the thunderstorm event estimated LoD, considering reprojection error, is almost two-fold compared to the LoD taking the references into account (22 and 13 mm, respectively), whereas for the rainfall simulation differences between both LoD are small. In this study the LoD considering the distances between the DEM of the first and last frame and the independent reference sources (multi-image reference) are used for change detection with DoDs due to their higher reliability. On the one hand, the compared point clouds inherit the advantage of even point distributions covering the entire area of interest, which is in contrast to a few single depicted point locations in the case of CPs. And on the other hand, the point clouds are gathered by a different data source and are thus, compared to the GCPs, not influenced by the time-lapse approach. The multi-image reference is preferred for both experiments to ensure measured changes are indeed surface changes. The building floor assessment could also be considered for the LoD calculation. But due to the artificial characteristic of the object, which is very different to a rough

Table II. Comparison of the three-dimensional (3D) reconstructed surface at simulated field and plot application (Experiments 1 and 2) to a total station reference grid

	Distorted		Undistorted images	
	Mean	Standard deviation	Mean	Standard deviation
Field simulation (mm)	−1.7	3.7	1.9	1.8
Field simulation with ICP (mm)	0.0	2.8	0.0	1.7
Plot simulation (mm)	−1.5	3.3	−3.1	1.6
Plot simulation with ICP (mm)	0.0	1.4	0.0	1.1

Table III. Accuracies achieved at the field scale [unmanned aerial vehicle (UAV)] and at the plot scale (terrestrial) assessed with independent UAV data and terrestrial structure-from-motion (SfM) data, respectively, as reference

	UAV (for Experiment 1)				Terrestrial (for Experiment 2)			
	13 May 2016		28 May 2016		Pre-rainfall		Post-rainfall	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Without ICP (mm)	−3	6	5	10	−0.5	2.11	−0.1	1.39
With ICP (mm)	0	6	0	8	0	1.93	0	1.21

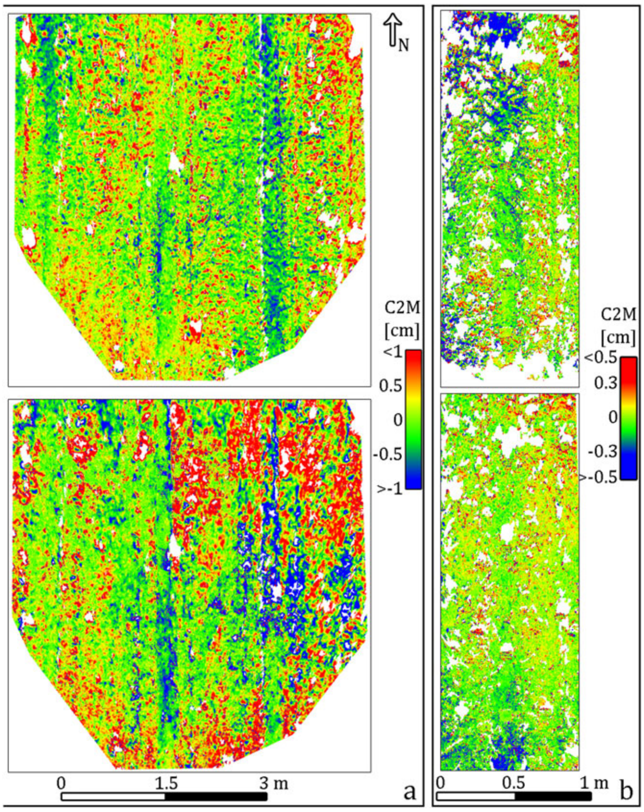

Figure 6. (a) C2M point distances between the meshed unmanned aerial vehicle (UAV) image derived structure-from-motion (SfM) point cloud and the reconstructed time-lapse model – prior (upper maps) and posterior (lower maps) the actual four-dimensional (4D) observation at field scale (b) C2M point distances between the meshed terrestrial image derived SfM point cloud and the reconstructed time-lapse model – prior (upper maps) and posterior (lower maps) the actual 4D observation at plot scale. Orthophotos of the investigated area of interest is displayed in Figure 9 (Experiment 1) and Figure 12 (Experiment 2). [Colour figure can be viewed at wileyonlinelibrary.com]

Table IV. Averaged three-dimensional (3D) reconstruction error of the digital elevation model (DEM) of each frame and corresponding level of detection (LoD) due to error propagation (reprojection error) is depicted

	Experiment	GCP	CP	Weighted error	LoD (0.9)
Reprojection	1	4.3	15.8	12	22
error (mm)	2	1.3	2.3	2	3.6
C2M	1	—	—	7	13
error (mm)	2	—	—	1.6	2.8

Note: Additionally, LoD is displayed using the error values from the independent reference consideration in Table III (C2M error).

soil surface, this reference investigation is solely used for best case scenario error estimation.

The estimation of the LoD is an important course of action to distinguish actual surface changes from noise with a profound statistical approach. However, in this study the LoD is considered to be too pessimistic because the raw DoD (i.e. subtraction of two DEMs) is not used for change detection but a DoD filtered along the temporal domain to diminish random error (Figure 7), which is possible due to the high frequency of data acquisition. Nevertheless, the LoD is implemented due to a missing alternative to assess the actual impact of the 4D filtering at error reduction.

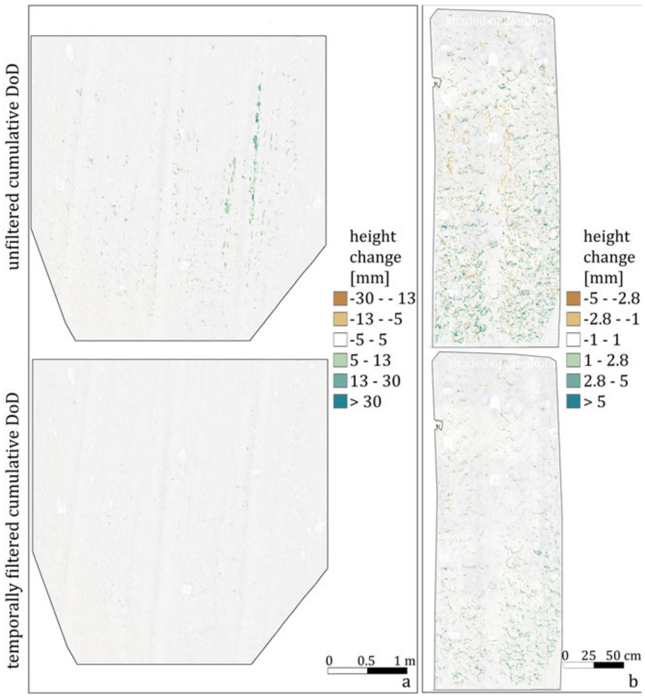

Figure 7. Illustration of the effectiveness of temporal filtering. For both experiments a digital elevation model (DEM) of difference (DoD) is displayed for a time interval when no changes were expected (Experiment 1 first nine minutes and Experiment 2 first six minutes of data capturing). Filtering results in detected height changes below level of detection (LoD) (13 mm Experiment 1 and 2.8 mm Experiment 2). [Colour figure can be viewed at wileyonlinelibrary.com]

Results of experiment 1 - Four-dimensional (4D) reconstruction of geomorphic changes at field scale

Rainfall captured from RADAR data occurs almost the entire observation period with a typical Weibull-density function, with a period of low intensities at the beginning, strongly increasing intensities after 55 minutes and slightly decreasing intensities after 65 minutes (Figure 8). The assessment of the 4D reconstruction data reveals that height changes in the DoDs during the event are mostly due to runoff changes rather than an actual increase or decrease of the soil surface. However, these results can be of significance, as well. In the subsequent description of the surface changes additional information is considered, i.e. measured runoff, rainfall characteristic, and data quality that is expressed counting no data cells (Figure 8). In the following, soil surface variations are described in different sections corresponding to periods of significant events or obvious changes:

(1) Period of little change: At the beginning of the event intensity is low (0.1 mm/min) and not sufficient for runoff generation (Figure 9, image 1). Positive surface changes, occur mainly within the interspaces of the wheel tracks (Figure 9, DoD 1).
(2) First marginal surface increase: After 45 minutes of rainfall a clear surface heightening is detectable (Figure 9, DoD 2), which coincides with slightly delayed increase of rainfall intensity. The temporal shift between surface and precipitation change is likely caused by the low spatial resolution of the radar data. The passing of the local rainfall cell could have occurred earlier at the area of interest. From the visual point of view, no obvious changes are detectable at the plot.

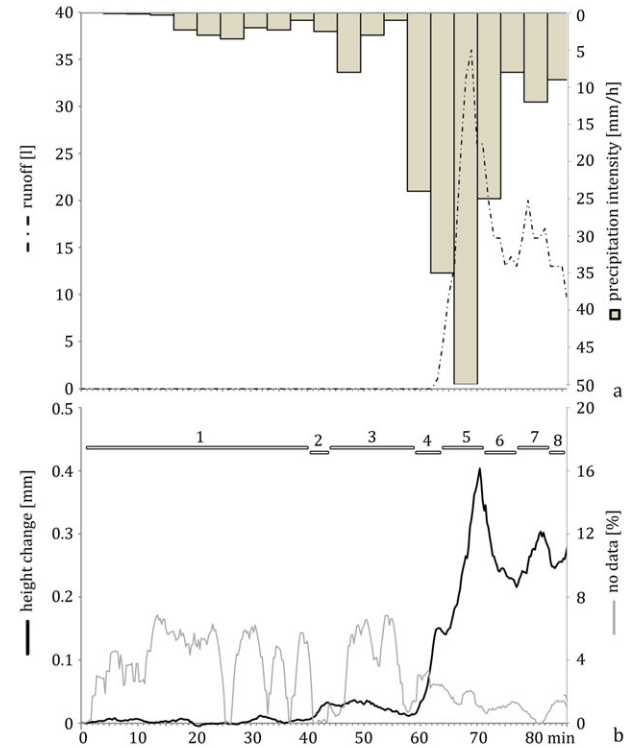

Figure 8. Four-dimensional (4D) reconstruction of the soil surface change at field scale (Experiment 1) during a thunderstorm event considering a level of detection (LoD) of 1.3 cm. (a) Rainfall intensity from RADAR data and tipping bucket measured runoff. (b) Area averaged elevation change and amount of no data cells. Numbers represent different periods of observed change and are described in detail in the text. [Colour figure can be viewed at wileyonlinelibrary.com]

(3) Marginal surface decrease: After surface heightening soil surface decreases slightly (Figure 9, DoD 3), lasting however only a few minutes.
(4) Captured runoff: Formation of runoff is captured by DoD (Figure 9, image 4) and also by tipping bucket measurement at the bottom of the investigated field plot (approximately 5 m away from the lower end of the surveyed area).
(5) Maximum surface occlusion by water: The measured surface height increases to a tipping point (> 4 mm), when maximum areal coverage of the surface by water due to runoff and local retention spots is reached (Figure 9, image 5). In contrast to phase 4, the trough measured runoff peak is reached earlier then the observed changes in the images due to water remaining in local retentions.
(6) Runoff decrease: When rainfall intensity decreases, surface height decreases, as well. However, the surface is still significantly higher (Figure 9, DoD 6).
(7) Repeated runoff increase: Afterwards, runoff and thus reconstructed surface heightens one more time due to increasing rainfall intensity, which can be seen in the images (Figure 9, image 7).
(8) Cumulative surface change of the entire observed event: The final DoD (Figure 9, DoD 8) summarizes the captured changes occurring during the entire thunderstorm event. Soil surface changes mainly occur due to sediment deposition in the wheel tracks (Figure 10).

Results of experiment 2 - Four-dimensional (4D) reconstruction of geomorphic changes at plot scale

Rainfall intensity remained constant during the entire observation (1 mm/min) and sediment yield was measured besides runoff (Figure 11). Experiment 2 reacts slightly different during the

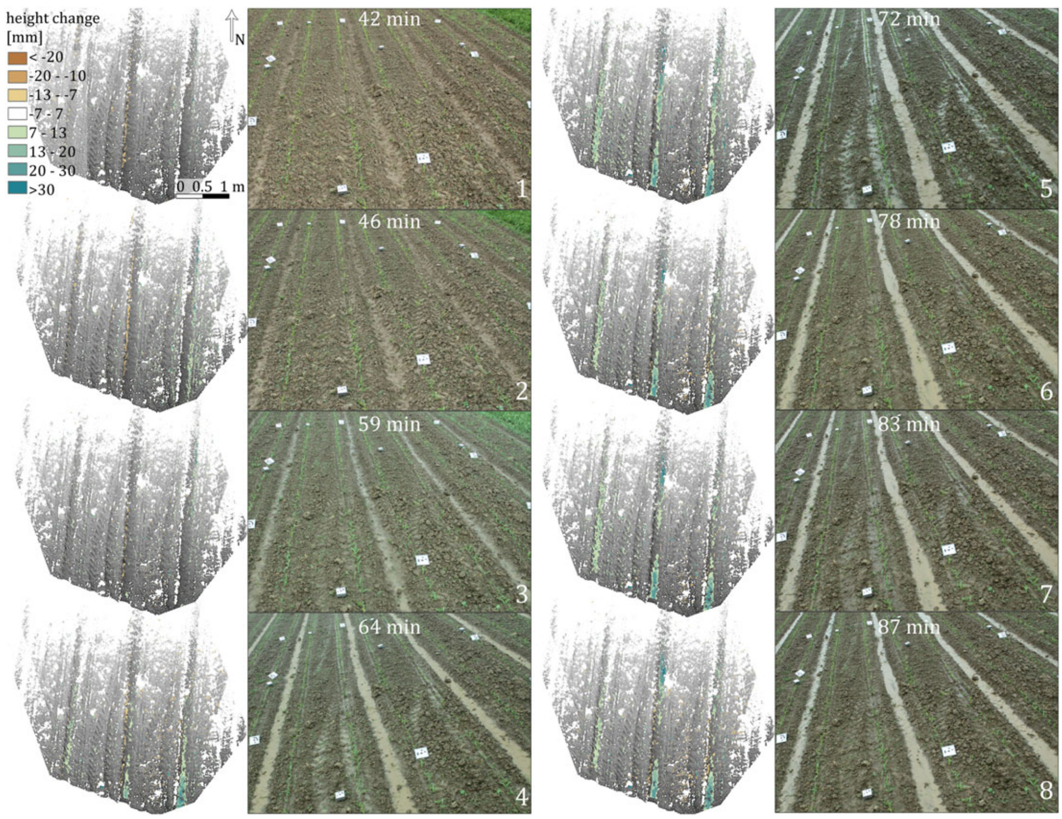

Figure 9. Digital elevation models (DEMs) of difference (DoDs) and corresponding images displaying significant changes of the observed event during Experiment 1 (field scale). DoDs are underlain by hillshade raster of the first DEM. Thus, grey areas indicate no change occurred and white areas represent no data. [Colour figure can be viewed at wileyonlinelibrary.com]

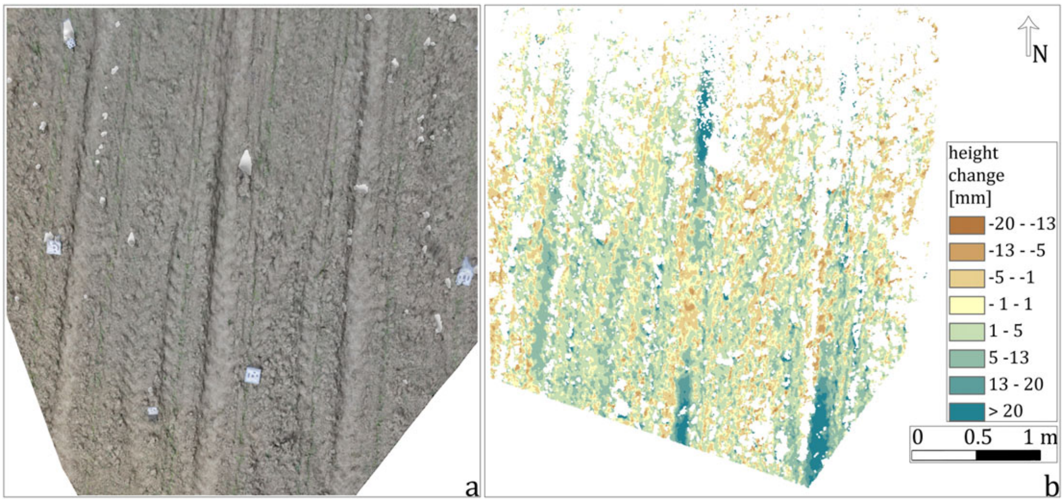

Figure 10. Cumulative surface change during the thunderstorm event at the investigated field plot (Experiment 1). (a) Shaded orthophoto of the area of interest. (b) Digital elevation model (DEM) of difference (DoD) is temporally filtered allowing for illustration of changes beneath the level of detection (LoD) of 1.3 cm. A period with limited surface runoff is chosen for the cumulative change display or else runoff would mask changes especially within the wheel tracks. [Colour figure can be viewed at wileyonlinelibrary.com]

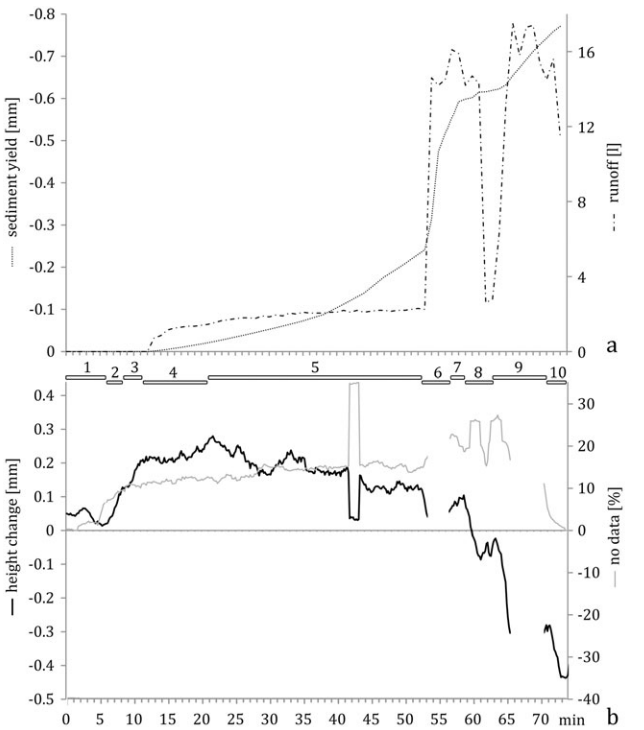

Figure 11. Four-dimensional (4D) reconstruction of soil surface changes at plot scale (Experiment 2) during a rainfall simulation. (a) Sediment yield and measured runoff. (b) Area averaged elevation change and amount of no data cells. Sediment yield and runoff are displayed, also.

rainfall simulation than Experiment 1 because obvious elevation changes occur within the first minutes of constant rainfall presumably due to higher precipitation intensities:

(1) Pre-rainfall: The timeline of soil surface changes reveals some height deviations before rainfall starts and thus has to be attributed to measured external factors, e.g. due to vegetation movement, rather than actual surface changes.
(2) Surface increase: Precipitation begins after five minutes of observation followed by an increase of the surface, especially in the lower half of the plot (Figure 12, DoD 2),
(3) Captured runoff: Developing runoff causes significant positive surface changes (Figure 12, image 3). Although surface area is still heightening, first runoff and suspended sediments are measured at the plot outlet after eight minutes of rainfall.
(4) Enduring surface heightening: The reconstructed DEM heights keep increasing, amongst others due to growing water levels and additional formation of local water retention areas (merging ponds), until a peak is reached after 21 minutes (Figure 12, image 4).
(5) Slight surface decrease: An almost steady state of slow surface decrease lasts for about 30 minutes (Figure 12, image 5). Thereby, the trend of soil loss measured by the DoDs corresponds with measured trend of sediment losses at the plot outlet. Interrill erosion is the captured process because changes at rill positions are not measured by SfM due to masking by runoff.
(6) Increased failure of 3D reconstruction due to increased turbulent runoff: After about 50 minutes of rainfall additional runoff is released at the micro-plot. This added runoff causes a failure of surface reconstruction during the first minutes of high runoff, which is mainly turbulent. Solely, at the beginning of runoff increase DEMs are calculated (Figure 12, image 6) until the moment (54 minutes in Figure 11) turbulent runoff covers too large an area. Increasing no-data values of the first minutes of added runoff are caused by synchronization problems which limit suitable capture of standing waves, which should be solved by automatic synchronization approaches in future studies.
(7) Steep runoff reconstruction: When turbulent runoff decreases in return for laminar flow DEMs are available again (Figure 12, image 7). In these surface models high water levels are measured, which becomes obvious after added runoff has been turned off because then soil heights decrease significantly.
(8) Post-added runoff surface: When the reflux experiment is ended no data values decrease to the level before reflux started. Generally, the impact of the additional runoff for soil erosion is measureable because the DoD remains distinctively below the pre-added runoff phase (Figure 12, DoD 8) and surface changes drop almost to the initial average surface height. Measured sediment yield increases significantly. Until this point more than 6 kg of sediment is lost. However, this does not coincide with

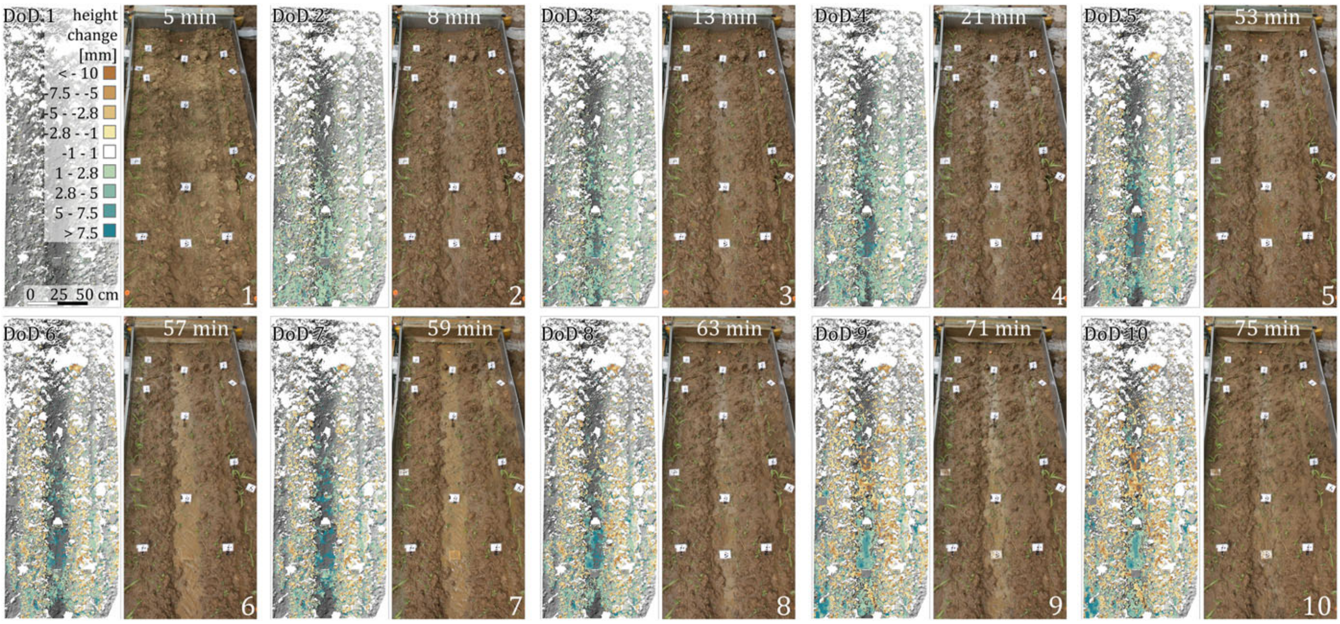

Figure 12. Digital elevation models (DEMs) of difference (DoDs) and corresponding images of significant events of soil surface change at the observed rainfall simulation plot (Experiment 2). DoDs are underlain by hillshade raster of the first DEM. Thus, grey areas indicate no change occurred and white areas represent no data. [Colour figure can be viewed at wileyonlinelibrary.com]

the cumulative DoD because the surface is still covered by runoff and local retention areas due to continuing rainfall and infiltration excess (Figure 12, image 8).

(9) Remaining retention areas: During the second increased reflux erosion processes resume and after the second time of added runoff the denudation process induced by higher runoff rates becomes more pronounced (Figure 12, DoD 9). However, the rainfall simulator is still running and therefore small runoff and local retention spots are still present.

(10) Cumulative changes of post-rainfall simulation: The final DEM height decreases significantly after the rainfall simulation has been terminated (70 minutes, Figure 11) because then the entire eroded rill becomes visible after runoff and local ponds are infiltrated and/or evaporated (Figure 12, image 10).

DoD measured denudation amounts up to 0.45 mm. However, total loss is estimated about 0.6 mm by DoD application considering a detected initial average uplift by hydrostatic pressure and/or swelling of approximately 0.15 mm.

Discussion

Four-dimensional (4D) reconstruction of geomorphic changes at field scale (experiment 1)

During the first period of the rainfall event, when runoff has not formed yet, elevated areas are mostly found within the depressions caused by ponding processes (Chu *et al.*, 2013). Areas of continuing surface elevation increases are caused by on-going ponding but may also be a result of swelling and/or hydrostatic effects. Furthermore, splash transport increases, which results in deposition areas at the easternmost wheel track.

Regions of surface lowering are a result of an increased destruction of soil aggregates due to the intensification of rainfall. If raindrops are hitting the bare soil, particles are detached by falling droplets (Le Bissonnais, 1996). Downsized particles can be moved up to a few decimetres by splash (Poesen and Savat, 1981; Marzen *et al.*, 2015). However, this possible impact is masked quickly due to the following formation of runoff, which causes a distinct increase of the reconstructed surface. Initial flow is filling and spilling micro-depressions, which consequently result in a changing flow pattern (Zhao *et al.*, 2016). Infiltration excess overland flow is relevant because it can detach and transport soil particles (Auerswald, 1998).

Changes in the wheel tracks are reconstructed with higher reliability due to larger magnitudes of change revealing areas of accumulation at locations where the tracks are wider alternating with areas of no change or erosion where the tracks become narrower (Figure 10) and thus runoff accelerates leading to increased transport capacity (Alberts *et al.*, 1980). Examining images from the event when runoff was largest identifies pool and ripple sequences, displayed by laminar and turbulent segments, respectively. However, as long as runoff and pools are present at the surface, change detection due to erosion and accumulation is difficult due to the non-distinguishing reconstruction of runoff and soil surface. This needs further consideration in future studies.

Nevertheless, due to the high temporal resolution the visibility of the runoff pattern and its changes (Figure 9, images 3–5) can be important too, e.g. to successfully explain the complexity of soil erosion processes (Quinton, 2004) and clearly highlight the time-lapse data potential to observe inner event variability.

Four-dimensional (4D) reconstruction of geomorphic changes at plot scale (experiment 2)

Elevating surfaces during the first period of rainfall simulation is mainly caused by swelling or hydrostatic uplift because no runoff or local ponds have formed, yet, that could cause additional height offsets. Hydrostatic uplift is more plausible because the surface heightening is more pronounced in the lower half of the plot, where pore water is drained gravitationally towards the lower border of the plot (Kaiser *et al.*, submitted for publication). Surface lowering caused by local erosion, mostly detected at edges of larger aggregates, occurs due to aggregate breakdown and splash transport, which starts to exceed uplift because of constant sediment loss due to high runoff rates. Continuing runoff concentration in morphologically or structurally pre-shaped areas results in linear runoff and erosion features (rill erosion). Retreating rills are competing during an event or a cascade of events in a way that only the most successful become continuous (Favis-Mortlock, 1998). This effect could rudimentary be detected along the left plot side (Figure 12, images 4–10).

Clear distinction between different processes and thus corresponding magnitudes of influence at the surface change is difficult with this study design. For instance, swelling amounts could be masked by runoff and prevailing denudation processes. Furthermore, surface observations stopped immediately after the rainfall simulation has ended. Thus, correction of height changes, considering drying and shrinking to the initial height of the surface, was not possible. And even if the dry surface could have been considered, other changes, e.g. compaction due to raindrop impact are also difficult to assess when comparing DoDs and measured sediment yield (Hänsel *et al.*, 2016; Kaiser *et al.*, submitted for publication).

Another aspect that needs to be considered is the fact that elevation changes are detected with highest reliability in the lower half of the plot close to the cameras. This area is closer to the local erosion basis and thus lower erosion is expected compared to the upper plot area, which is captured only partly. Measured sediment yield is not affected by this difference. Generally, the reconstructed plot area contains large data gaps and cumulative changes averaged for the entire plot area can thus underestimate actual surface changes (Figure 13). Although, errors of the sediment measurement at the plot outlet are also relevant, for instance considering changes in bulk density (Hänsel *et al.*, 2016). It is still not certain to which amount bulk density at the upper soil layer changes during splash impact, swelling, compaction by water, and deposition of sediment (e.g. Flemming and Dalfontaine, 2016; Kaiser

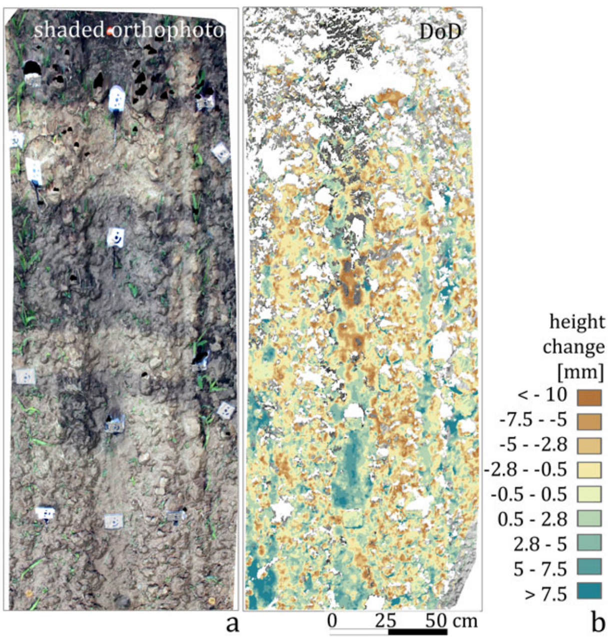

Figure 13. Cumulative surface change during the rainfall simulation at the investigated small plot (Experiment 2). (a) Shaded orthophoto. (b) Digital elevation model (DEM) of difference (DoD) is temporally filtered allowing for illustration of changes beneath the level of detection (LoD) of 2.8 mm. A frame after the rainfall simulation termination is chosen to ensure no more water at the surface. [Colour figure can be viewed at wileyonlinelibrary.com]

et al., submitted for publication). Under these circumstances the measured sediment yield of 0.8 mm matches quite well to the changes detected with 4D SfM reconstruction (0.6 mm).

Significant decrease of surface height was only measured during and after periods of high runoff rates. Thus, the presented method could, besides the mentioned shortcomings, serve as a missing link to quantify intra-storm process alterations, whose relevance has already been discussed (Favis-Mortlock *et al.*, 2000; Darboux *et al.*, 2001; Mohamadi and Kavian, 2015; Zhao et al., 2016).

Challenges and Perspectives

A series of technical aspects are currently limiting our ability to reconstruct geomorphic changes with great detail, including the existence of shadow areas, the exact synchronization of the cameras and the existence of runoff impeding surface geometry reconstruction, as follows:

(a) Data quality has to be considered carefully at the plot scale during the rainfall simulation, especially regarding no data values, when evaluating the soil surface changes over time. Increasing number of no data raster cells causes a significant drop of the DoD value. Especially, if runoff reconstruction fails, areas outside the rill, which are mostly characterized by aggregate break down and erosion of fine substrate, gain more weight.

(b) Camera synchronization: this is a key aspect for the success of runoff reconstruction and for reducing the number of no data cells. On the one hand, fast changes might not be reconstructed if the captured scene is too different in the three images when pictures are not synchronous. On the other hand, runoff can only be reconstructed during relatively steady conditions. This was particularly obvious at the plot scale. During the whole rainfall simulation experiment the number of no data values increases slowly, which is caused by slowly increasing water covered areas due to increasing runoff. The areas capped by moving water are difficult to reconstruct due to insufficient camera synchronization, which is not the case for the local ponds with still water surfaces. However, when the runoff reflux experiment is started no data increases significantly resulting from large areas covered by turbulent flow (Figure 12, image 6). Only when turbulent runoff eases off a few minutes after reflux has started more runoff becomes reconstructed (Figure 12, image 7). To increase camera synchronization the utilization of either wireless (WLAN), wired triggers or global positioning system (GPS) time are potential approaches.

(c) The reconstruction of the runoff causes a general difficulty if the surface changes are to be evaluated because the soil surface beneath the water layer is not assessable. Furthermore, when runoff occurs increasing positive cumulative DoD values are no more solely approachable as a surface heightening. Thus, in future application other sensors are needed to be integrated to allow for a reliable and automatic distinction between surface and runoff. Thereby, thermal imaging could be a potential solution due to the high radiometric resolution (0.1°C) enabling the separation of slightly different temperate materials (Figure 14).

Only three cameras are used in this study. However, more cameras are a prerequisite if shadow effects are to be avoided and larger areas are to be captured. Also, camera setup at varying distances are relevant to decrease the decreasing data

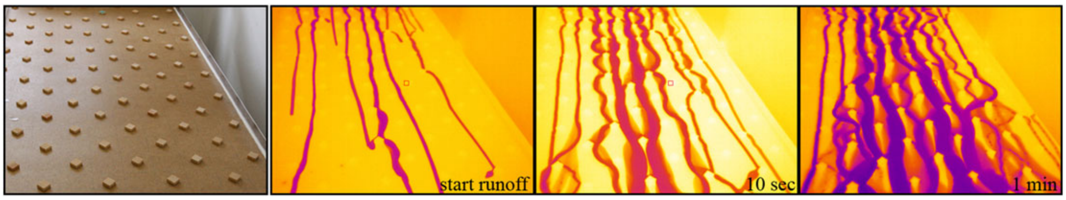

Figure 14. Example for the application of additional sensors to automatically distinguish between soil surface and runoff. [Colour figure can be viewed at wileyonlinelibrary.com]

quality with increasing distance to the sensors, especially for oblique viewing angles.

In geomorphic research the main focus is put on seasonal dynamics and on space–time interaction. The presented continuous monitoring approach can support *in situ* measurements on high temporal resolutions. Potential utilizations could be in volcanic studies, as already demonstrated by James and Robson (2014b). Other applications are the monitoring of landslides and rock falls, allowing for change detection of fast occurring as well as longer term events. In soil erosion studies also larger forms are measureable, enabling the observation of rill and gully formation during single rainfall events. In hydrological studies continuous 3D monitoring can be of importance to monitor bank erosion and river cross-section changes, which are relevant for discharge estimations. In coastal geomorphology dune formation and cliff erosion can be quantified. When the encountered challenges for time-lapse SfM photogrammetry are considered an implementation for a significantly broader field of applications is possible, i.e. all short- to mid-term changes with sufficient magnitudes of alterations to overcome the LoD at the corresponding scales.

Conclusion

In this study, an approach to quantify surface changes with high spatial and temporal resolution utilizing time-lapse photogrammetry is presented. Three synchronized SLR cameras are used to acquire images in sub-minute intervals. The application of 4D reconstructed surfaces allows for monitoring changes also during the event. Furthermore, significant changes during the observation period can be assessed individually in high detail consulting each corresponding image and DoD.

The soil surface was measured during a thunderstorm event at field scale revealing erosion and deposition sequences due to pool-ripple-systems in the wheel tracks. During a rainfall simulation at plot scale changes are detected with millimetre accuracy allowing for the observation of hydrostatic and/or swelling effects as well as interrill erosion due to aggregate break down. Moreover, pool-ripple-sequences are observable again in the rill.

In future studies the runoff needs to be considered carefully to separate reconstructed overland flow from actual surface changes. However, if this factor is accounted for, runoff depth and its changes are measurable allowing for new insights regarding sedimentological and hydrological connectivity.

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References

Abellán A, Jaboyedoff M, Oppikofer T, Vilaplana JM. 2009. Detection of millimetric deformation using a terrestrial laser scanner: experiment and application to a rockfall event. *Natural Hazards and Earth System Sciences* **9**(2): 365–372.

Arce GR. 2005. *Nonlinear Signal Processing: A Statistical Approach*. Wiley: Hoboken, NJ.

Alberts EE, Moldenhauer WC, Foster GR. 1980. Soil Aggregates and primary particles transported in rill and interrill flow. *Soil Science Society of America Journal* **44**: 590–595.

Auerswald K. 1998. Bodenerosion durch Wasser. In *Bodenerosion: Analyse und Bilanz eines Umweltproblems*, Richter G (ed). Wissenschaftliche Buchgesellschaft: Darmstadt; 264.

Bradski G, Kaehler A. 2008. *Learning OpenCV*. O'Reilly: Sebastopol; 556.

Bracken LJ, Turnbull L, Wainwright J, Bogaart P. 2015. State of science sediment connectivity: a framework for understanding sediment transfer at multiple scales. *Earth Surface Processes and Landforms* **40**: 177–188.

Brasington J, Langham J, Rumsby B. 2003. Methodological sensitivity of morphometric estimates of coarse fluvial sediment transport. *Geomorphology* **53**(3–4): 299–316.

Carrivick J, Smith M, Quincey D. 2016. *Structure from Motion in the Geosciences*. Chichester: Wiley-Blackwell; 208.

Chu X, Yang J, Chi Y, Zhang J. 2013. Dynamic puddle delineation and modeling of puddle-to-puddle filling-spilling-merging-splitting overland flow processes. *Water Resources Research* **49**(6): 3825–3829.

Darboux F, Davy P, Gascuel-Odoux C, Huang C. 2001. Evolution of soil surface roughness and flowpath connectivity in overland flow experiments. *Catena* **46**: 125–139.

Dietrich JT. 2016. Riverscape mapping with helicopter-based structure-from-motion photogrammetry. *Geomorphology* **252**: 144–157.

Eitel J, Höfle B, Vierling L, Abellán A, Asner G, Deems J, Glennie C, Joerg P, LeWinter A, Magney T. 2016. Beyond 3D: the new spectrum of LiDAR applications for earth and ecological sciences. *Remote Sensing of Environment* **186**: 372–392.

Eltner A, Baumgart P. 2015. Accuracy constraints of terrestrial Lidar data for soil erosion measurement: Application to a Mediterranean field plot. *Geomorphology* **245**: 243–254. https://doi.org/10.1016/j.geomorph.2015.06.008

Eltner A, Baumgart P, Maas H-G, Faust D. 2015. Multi-temporal UAV data for automatic measurement of rill and interrill erosion on loess soil. *Earth Surface Processes and Landforms* **40**(6): 741–755.

Eltner A, Kaiser A, Castillo C, Rock G, Neugirg F, Abellan A. 2016a. Image-based surface reconstruction in geomorphometry – merits, limits and developments. *Earth Surface Dynamics* **4**: 359–389.

Eltner A, Schneider D, Maas H. 2016b. Integrated processing of high resolution topographic data for soil erosion assessment considering data acquisition schemes and surface properties. *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **XLI-B5**: 813–819. https://doi.org/10.5194/isprs-archives-XLI-B5-813-2016

Eltner A, Mulsow C, Maas H-G. 2013. Quantitative measurement of soil erosion from TLS and UAV data. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **XL-1/W2**: 119–124.

Eltner A, Schneider D. 2015. Analysis of different methods for 3D reconstruction of natural surfaces from parallel-axes UAV images. *The Photogrammetric Record* **30**(151): 279–299.

Favis-Mortlock D. 1998. A self-organizing dynamic systems approach to the simulation of rill initiation and development on hillslopes. *Computers & Geosciences* **24**(4): 353–372.

Favis-Mortlock DT, Boardman J, Parsons AJ, Lascelles B. 2000. Emergence and erosion: a model for rill initiation and development. *Hydrological Processes* **14**: 2173–2205.

Flemming B, Delafontaine K. 2016. Mass Physical Sand Properties. In *Enyclopedia of Estuaries*, Kennish MJ (ed). Springer: Berlin; 419–432.

Gabrieli F, Corain L, Vettore L. 2016. A low-cost landslide displacement activity assessment from time-lapse photogrammetry and rainfall data: application to the Tessina landslide site. *Geomorphology* **269**: 56–74.

Grün A. 2012. Development and status of image matching in photogrammetry. *The Photogrammetric Record* **27**(137): 36–57.

Hänsel P, Schindewolf M, Eltner A, Kaiser A, Schmidt J. 2016. Feasibility of high-resolution soil erosion measurements by means of rainfall simulations and SfM photogrammetry. *Hydrology* **3**(4): 38.

James MR, Robson S. 2012. Straightforward reconstruction of 3D surfaces and topography with a camera: accuracy and geoscience application. *Journal of Geophysical Research* **117** F03017.

James MR, Robson S. 2014a. Sequential digital elevation models of active lava flows from ground-based stereo time-lapse imagery. *ISPRS Journal of Photogrammetry and Remote Sensing* **97**: 160–170.

James MR, Robson S. 2014b. Mitigating systematic error in topographic models derived from UAV and ground-based image networks. *Earth Surface Processes and Landforms* **39**: 1413–1420.

Kaiser A, Eltner A, Erhardt A, Müller C. Submitted for publication. Addressing uncertainties in soil erosion research: non-erosive soil surface changes in multi-temporal high resolution topography data across scales. *Land Degradation and Development*.

Kaiser A, Neugirg F, Rock G, Müller C, Haas F, Ries J, Schmidt J. 2014. Small-Scale Surface Reconstruction and Volume Calculation of Soil Erosion in Complex Moroccan Gully Morphology Using Structure from Motion. *Remote Sensing* **6**: 7050–7080.

Kinell PIA. 2016. A review of the design and operation of runoff and soil loss plots. *Catena* **145**: 257–265.

Kramer N, Wohl E. 2014. Estimating fluvial wood discharge using time-lapse photography with varying sampling intervals. *Earth Surface Processes and Landforms* **852**: 844–852.

Kromer R, Abellán A, Hutchinson D, Lato M, Edwards T, Jaboyedoff M. 2015. A 4D filtering and calibration technique for small-scale point cloud change detection with a terrestrial laser scanner. *Remote Sensing* **7**(10): 13029–13052.

Lane SN, Westaway RM, Murray HD. 2003. Estimation of erosion and deposition volumes in a large, gravel-bed, braided river using synoptic remote sensing. *Earth Surface Processes and Landforms* **28**: 249–271.

Le Le Bissonnais Y. 1996. Aggregate stability and assessment of soil crustability and erodibility: I. *Theory and methodology. European Journal of Soil Science* **47**: 425–437.

Luhmann T, Robson S, Kyle S, Boehm J. 2014. *Close-range Photogrammetry and 3-D Imaging*, second edn. Berlin: De Gruyter; 683.

Maas H-G, Casassa G, Schneider D, Schwalbe E, Wendt A. 2013. Photogrammetric techniques for the determination of spatio-temporal velocity fields at Glaciar San Rafael, Chile. *Photogrammetric Engineering & Remote Sensing* **3**: 299–306.

Martin-Brualla R, Gallup D, Seitz SM. 2015. Time-lapse mining from Internet photos. *Proceedings, IEEE International Conference on Computer Vision (ICCV)*, 13–16 December.

Marzen M, Iserloh T, Casper M, Ries JB. 2015. Quantification of particle detachment by rain splash and wind-driven rain splash. *Catena* **127**: 135–141.

Matsuoka N. 2014. Combining time-lapse photography and multisensor monitoring to understand frost creep dynamics in the Japanese Alps. *Permafrost and Periglacial Processes* **25**(2): 94–106.

Micheletti N, Chandler JH, Lane SN. 2014. Investigating the geomorphological potential of freely available and accessible structure-from-motion photogrammetry using a smartphone. *Earth Surface Processes and Landforms* **40**: 473–486.

Mohamadi MA, Kavian A. 2015. Effects of rainfall patterns on runoff and soil erosion in field plots. *International Soil and Water Conservation Research* **3**(4): 273–281.

Morgan R. 2005. *Soil Erosion and Conservation*, third edn. New Delhi: Blackwell Publishing; 304.

Mosbrucker A, Major J, Spicer K, Pitlick J. 2016. Camera system considerations for geomorphic applications of SfM photogrammetry. *Earth Surface Processes and Landforms* **42**(6): 969–986.

Nichols MH, Nearing M, Hernandez M, Polyakov VO. 2016. Monitoring channel head erosion processes in response to an artificially induced abrupt base level change using time-lapse photography. *Geomorphology* **265**: 107–116.

Parajka J, Haas P, Kirnbauer R, Jansa J, Blöschl G. 2012. Potential of time-lapse photography of snow for hydrological purposes at the small catchment scale. *Hydrological Processes* **26**(22): 3327–3337.

Passalacqua P, Belmont P, Staley DM, Simley JD, Arrowsmith JR, Bode CA et al. 2015. Analyzing high resolution topography for advancing the understanding of mass and energy transfer through landscapes: a review. *Earth-Science Reviews* **148**: 174–193.

Pesci M, Richter G. 1996. Löss. *Herkunft-Gliederung-Landschaften. Zeitschrift für Geomorphologie*, Suppl. 98: 391.

Poesen J, Savat J. 1981. Detachment and transportation of loose sediments by raindrop splash: Part II Detachability and transport ability measurements. *Catena* **8**(1): 19–41.

Prosdocimi M, Burguet M, Di Prima S, Sofia G, Terol E, Rodrigo Comino J, Cerdà A, Tarolli P. 2017. Rainfall simulation and structure-from-motion photogrammetry for the analysis of soil water erosion in Mediterranean vineyards. *Science of the Total Environment* **574**: 204–215.

Quinton J. 2004. Erosion and sediment transport. In *Environmental Modelling: Finding Simplicity in Complexity*, Wainwright J, Mulligan M (eds). Wiley-Blackwell: London; 187–196.

Schindewolf M, Schmidt J. 2012. Parameterization of the erosion 2D/3D soil erosion model using a small-scale rainfall simulator and upstream runoff simulation. *Catena* **91**: 47–55.

Schwalbe E, Koschitzki R, Maas H-G. 2016. Recognition of drainage tunnels during glacier lake outburst events from terrestrial image sequences. *ISPRS – International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **XLI-B8**: 537–543.

Smith MW, Carrivick JL, Quincey DJ. 2016. Structure from motion photogrammetry in physical geography. *Progress in Physical Geography* **40**(2): 247–275.

Snapir B, Hobbs S, Waine TW. 2014. Roughness measurements over an agricultural soil surface with structure from motion. *ISPRS Journal of Photogrammetry and Remote Sensing* **96**: 210–223.

Stöcker C, Eltner A, Karrasch P. 2015. Measuring gullies by synergetic application of UAV and close range photogrammetry — a case study from Andalusia, Spain. *Catena* **132**: 1–11.

Tarolli P. 2014. High-resolution topography for understanding Earth surface processes: opportunities and challenges. *Geomorphology* **216**: 295–312.

Zhao L, Huang C, Wu F. 2016. Effect of microrelief on water erosion and their changes during rainfall. *Earth Surface Processes and Landforms* **41**: 579–586.